

ALGEBRA I PRACTICE TEST



Answer all 6 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit.

25 Solve the equation below algebraically. *comeback*

pemdas

$$6 - \frac{2}{3}(x + 5) = 4x$$

→ we need to distribute first so it can be easier

$$6 - \frac{2}{3}x - \frac{10}{3} = 4x$$

↓

$$2.6 - \frac{2}{3}x = 4x \rightarrow 2.6 = 3\frac{1}{3}x$$

$$-\frac{2}{3}x \quad -\frac{2}{3}x \quad \frac{3\frac{1}{3}}{3\frac{1}{3}} \quad \frac{3\frac{1}{3}}{3\frac{1}{3}}x \quad \text{X} = .8$$

26 $F = 4x^2 - 3x + 1$ and $G = -2x^2 + 5$. Find the trinomial that represents H , if $H = 3G - F$.

$$3(-2x^2 + 5) - 4x^2 - 3x + 1$$

d

$$\begin{array}{ccccccc} -6x^2 & + & 15 & - & 4x^2 & - & 3x & + & 1 \\ 1 & & 2 & & 1 & & 1 & & 2 \end{array}$$

$$-6x^2 - 4x^2 - 3x = -10x^2 - 3x$$

$$15 + 1 = 16$$

$$H = -10x^2 - 3x + 16$$

comeback

27 Algebraically determine the zeros of the function $f(x) = x^2 - 4x + 3$.

$$x^2 - 4x + 3$$

$$0 = x^2 - 4x + 3$$

$$\begin{array}{r} -3 \qquad -3 \\ -3 = x^2 - 4x \\ +4x \qquad +4x \end{array}$$

$$\sqrt{4x-3} = \sqrt{x^2}$$
$$3.13 = x$$

- 28 In February 2021, the state of Texas saw one of the worst snowstorms in its history. Snowfall totals recorded in Dallas, TX are shown in the table below.

Dallas, TX	
Time	Snow (inches)
1 a.m.	1
3 a.m.	5
6 a.m.	11
12 noon	33
3 p.m.	36

Handwritten notes around the table:

- Left side: $+2H$, $+3H$, $+6H$, $+3H$ (grouped by a bracket)
- Right side: $+4$, $+6$, $+22$, $+3$ (grouped by a bracket)

Which of the following intervals had the greatest rate of snowfall, in inches per hour? 1 a.m. to 12 noon or 6 a.m. to 3 p.m. Justify your answer.

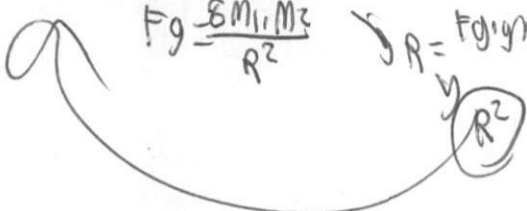
1 AM - 12 NOON = 32 Total inches of snow

6 AM - 3 PM = 25 Total inches of snow

From 1 AM to 12 NOON had the greatest Rate of snowfall because 32 inches fell but from 6 AM - 3 PM only 25 inches fell and that is a 7 inch difference so from 1 AM - 12 NOON had the greatest Rate of snowfall per hour.

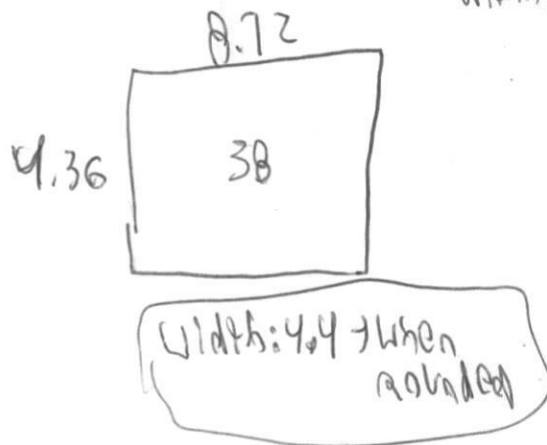
com esch

- 29 The gravitational force between two objects can be calculated using the formula $F_g = \frac{GM_1M_2}{r^2}$. In the formula, G is called the gravitational constant. The variables M_1 and M_2 represent the masses of the two objects between which the force is being measured. Lastly, r is the distance between the objects. Solve for the positive value of r , in terms of the other variables.

$$F_g = \frac{GM_1M_2}{r^2} \quad \rightarrow \quad R = \frac{F_g M_1 M_2}{R^2}$$


- 30 Four Seasons Landscaping is creating a rectangular flower bed at a customer's home. The width of the flower bed will be half of the length. The total area of the flower bed is 38 square feet. Write and solve an equation to determine the width of the flower bed to the nearest tenth of a foot.

width = half of the length



Part III

Answer all 4 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.

- 31 A couple planning a backyard wedding is researching different tent rental companies for their event. They contact two different companies. Buffalo Party Rental charges a \$50 set-up fee, and \$5 per square foot of tent space. Tent Genie charges a \$25 set-up fee, and \$6 per square foot of tent space.

Write an equation for Buffalo Party Rental that could be used to determine the total cost B , when x square feet of tent space is ordered. Write a second equation for Tent Genie that could be used to determine the total cost T , when x square feet of tent space is ordered.

two equations for the total cost

Buffalo Party: $50 + 5x = B$
Tent Genie: $25 + 6x = T$

combine

Determine algebraically and state the *minimum* square feet, rounded to the *nearest square foot*, of tent space that must be ordered for Buffalo Party Rental to be the cheaper option.

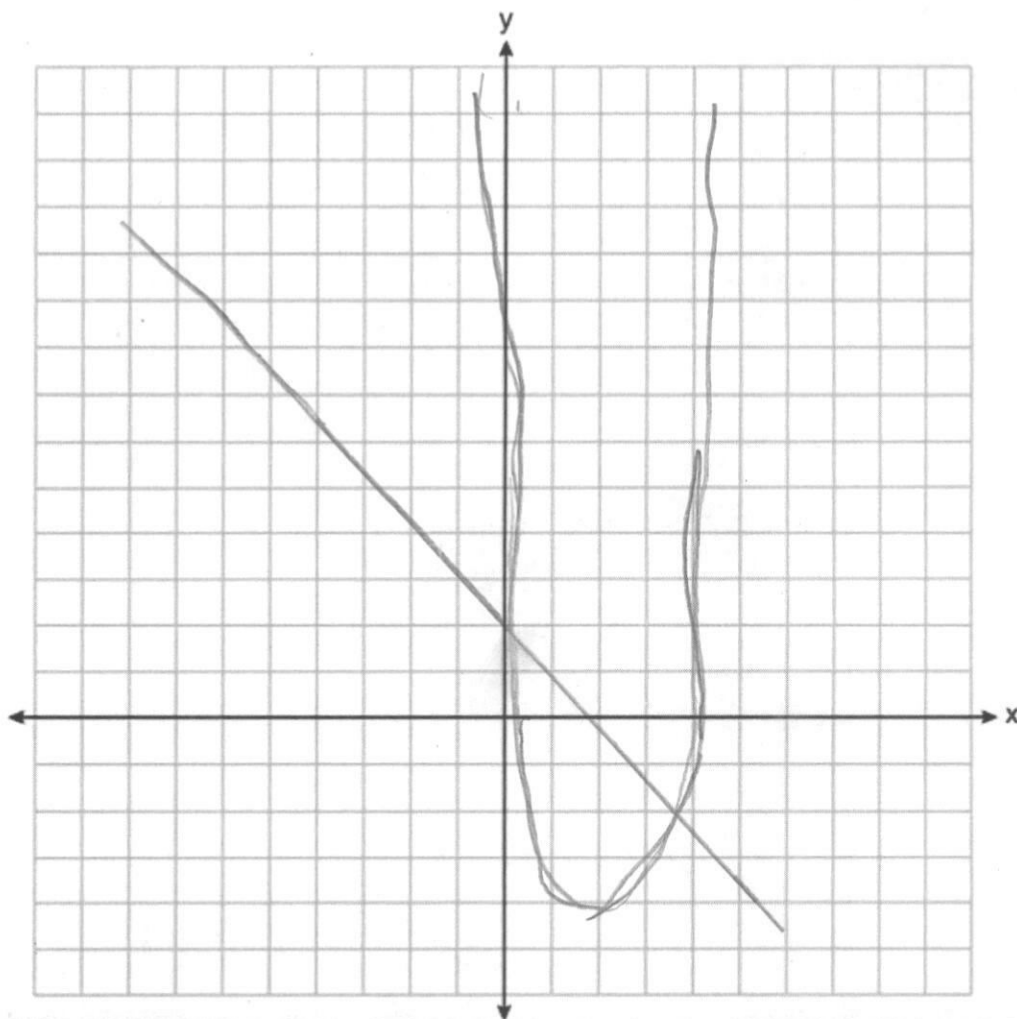
50 → square foot

→ $50 + 5(50) = B$

32 Graph the two functions on the coordinate plane below.

$$f(x) = 2x^2 - 8x + 3$$

$$g(x) = -2x + 3$$



Find all values of x for which $f(x) = g(x)$.

When $f(x) = g(x)$
The x are $(0, 3)$ and $(3, -3)$ — The x

- 33 The table below shows the number of hours ten different scholars at Marion P. Thomas Middle School spent studying and their scores on a recent history test.

Hours Spent Studying (x)	0	1	2	4	4	4	6	6	7	8
Test Scores (y)	35	40	46	65	67	70	82	88	82	95

Write the linear regression for this data set. Round all values to the nearest hundredth.

$$y = Ax + b$$

$$y = 7.79x + 34.27 = \text{Linear Regression}$$

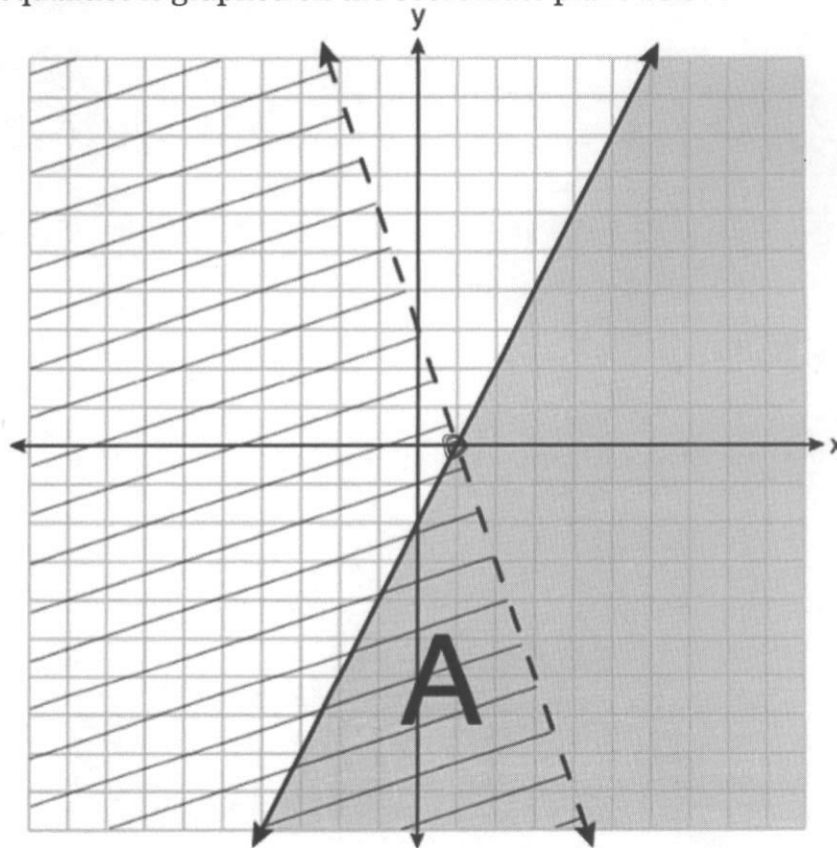
State the correlation coefficient rounded to the nearest hundredth.

$$\text{correlation coefficient} = .96$$

Explain what the correlation coefficient suggests in the context of the problem.

The correlation coefficient tells us the Average in context to the problem and also showing what it's near to.

34 A system of inequalities is graphed on the coordinate plane below.



* combine

Write the system of inequalities that is represented by the graph above.

~~1 > 0~~ 1 > 0

What does region A represent?

Where everything is going to, because the grey and the black lines all fit there which means is where A is mostly happening.

What does the gray region represent?

The Grey Region represents it is positive and that's why it shaded like that same meaning with the black lines.

Part IV

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil.

- 0349
- 35 The CoinStar machine at the local supermarket is broken. Instead of providing the exact number of each type of coin that is deposited, the machine can only display the total number of coins and total value. The machine shows 90 coins, with a value of \$17.55.

If Mr. Rogers only placed dimes and quarters in the machine, write a system of equations in two variables, or an equation in one variable that could be used to model this situation.

$$90d + 90q = 17.55 \quad \text{system of equations}$$

Using your equation or system of equations, algebraically determine the number of quarters Mr. Rogers deposited in the machine.

$$a \text{ quarter} = 0.25 \text{ cents}$$

$$90d + 90q = 17.55 \rightarrow 50 \text{ quarters}$$

Mr. Rogers' neighbor replaced each one of his dimes with a quarter. If he uses all of his coins, determine if Mr. Rogers would have enough money to buy a cardigan priced at \$20.98 if he must also pay an 8% sales tax. Justify your answer.

no He would not be able to buy a cardigan because with sales tax it would be more and would only get \$20 which is not enough even if he replaces the dimes with quarters.